

“IPL Dataset Analysis”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

Chavan Sakshi Sudhakar

❖ INTRODUCTION

The Indian Premier League (IPL) is one of the most popular T20 cricket leagues in the world, featuring top international and domestic players. Each season consists of multiple matches played across different venues, where teams compete based on batting strength, bowling performance, and strategic decisions.

The dataset used in this project contains match-level information such as teams, match winners, toss decisions, scores, top scorers, best bowlers, player of the match, and venues. This dataset provides a valuable opportunity to analyze real-world cricket data and understand the factors influencing match outcomes.

Exploratory Data Analysis (EDA) is a crucial step in data science that helps in understanding the structure of data, identifying patterns, and extracting meaningful insights through statistical methods and visualizations. In this project, EDA techniques are applied to clean the dataset, explore relationships between variables, and visualize trends using graphs such as bar charts and histograms.

The main aim of this analysis is to identify key factors that affect match results, such as team performance, toss impact, batting strategy, player contributions, and match competitiveness. This project demonstrates how data analysis techniques can be applied to sports analytics to derive useful insights.

❖ OBJECTIVES

1. To understand and clean the IPL dataset
2. To identify top-performing teams
3. To analyze the impact of toss on match results
4. To compare batting first vs chasing performance
5. To evaluate batsmen and bowlers performance
6. To analyze Player of the Match trends
7. To study venues and match competitiveness

❖ METHODOLOGY

The methodology of this project follows a systematic approach to analyze the IPL match dataset using Exploratory Data Analysis (EDA) techniques. Initially, the dataset is collected and loaded into the Python environment using libraries such as Pandas, NumPy, Matplotlib, and Seaborn. In the data understanding phase, functions like `head()`, `info()`, and `describe()` are used to examine the structure of the dataset, identify data types, and obtain summary statistics. This step helps in gaining a clear understanding of the dataset before proceeding further.

In the data cleaning stage, missing values and inconsistencies are handled to improve data quality. Important columns such as match winner are checked for null values, and the margin column is converted into numeric format to allow proper analysis. Column names are standardized to avoid errors during processing. After cleaning the dataset, exploratory data analysis is performed to identify patterns and relationships among variables.

Various analyses are conducted, including team performance analysis to determine the most successful teams based on match wins, toss impact analysis to evaluate whether winning the toss affects match outcomes, and batting strategy analysis to compare the effectiveness of batting first versus chasing. Player performance is analyzed by identifying top scorers, highest individual scores, best bowlers, and players who received the Player of the Match award. Venue analysis is also performed to understand the distribution of matches across different stadiums.

Data visualization techniques are applied to present the findings in a clear and meaningful way. Bar charts are used to compare team wins and player performances, while histograms are used to analyze the distribution of winning margins and understand match competitiveness. Finally, the results obtained from the analysis are interpreted to draw meaningful conclusions about team performance, player contributions, and match dynamics. This systematic approach helps in extracting valuable insights from the dataset and demonstrates the importance of EDA in sports analytics.

❖ Data Analysis

1.To understand and clean the IPL dataset & To identify top-performing teams

Code:

```
# Import Libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import warnings

warnings.filterwarnings('ignore')

# Load Dataset

df=pd.read_csv("IPL.csv")#2022_ipl

# Clean column names (important)

df.columns = df.columns.str.strip().str.lower()

# Fix missing values (if any)

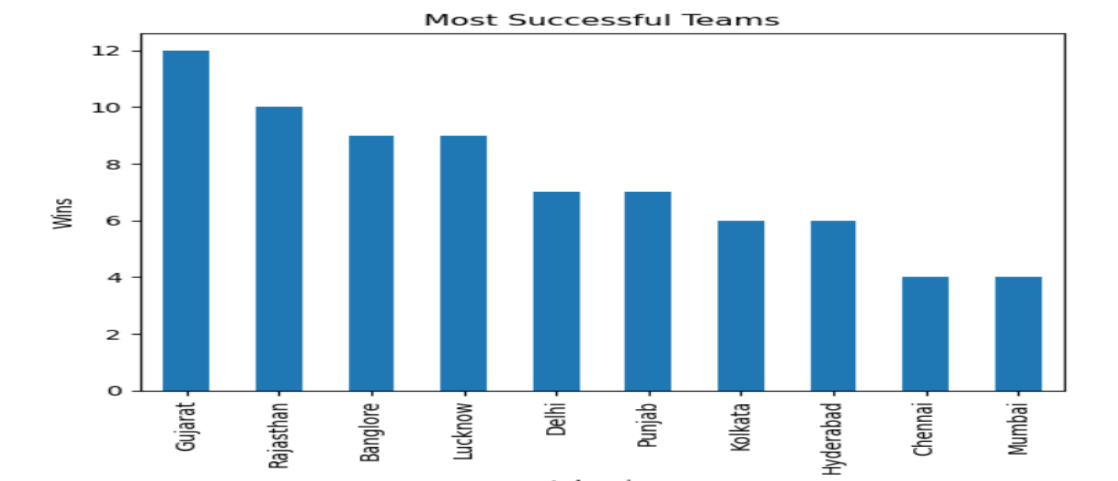
df = df.dropna(subset=["match_winner"])

top_teams =
df["match_winner"].value_counts()

print(top_teams)
```

Output:

match_winner	
Gujarat	12
Rajasthan	10
Banglore	9
Lucknow	9
Delhi	7
Punjab	7
Kolkata	6
Hyderabad	6
Chennai	4
Mumbai	4



Interpretation:

Gujarat is the most dominant team strong batting + bowling balance.
Team success in IPL depends on consistency across matches, not just star players.

2. To analyze the impact of toss on match results

Code:

```
toss_vs_match = df[df["toss_winner"]  
== df["match_winner"]]  
toss_win_rate = len(toss_vs_match) /  
len(df) * 100  
print("Toss Winner also won match  
(%):", toss_win_rate)
```

Output:

Toss Winner also won match (%): 48.64864864864865

Interpretation:

This is very close to 50% (random chance). Means toss has almost no advantage. Skill and performance dominate over luck”.

3. To compare batting first vs chasing performance

Code:

```
bat_first_wins = df[df["won_by"] == "Runs"].shape[0]
bat_second_wins = df[df["won_by"] == "Wickets"].shape[0]

print("Bat First Wins:", bat_first_wins)
print("Chasing Wins:", bat_second_wins)
```

Output:

```
Bat First Wins: 37
Chasing Wins: 37
```

Interpretation:

Perfect 50–50 distribution. No strategy is superior. Match outcome depends more on execution than strategy.

4. To evaluate batsmen and bowlers performance

Code:

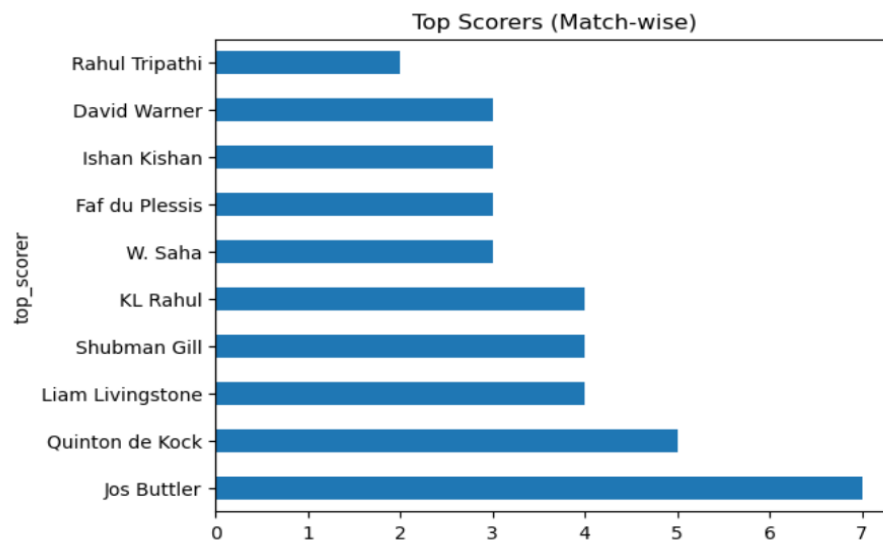
```
high_scores = df[["top_scorer", "highscore"]].sort_values(by="highscore", ascending=False).head(10)
print(high_scores)
best_bowlers = df["best_bowling"].value_counts().head(10)
print(best_bowlers)
```

Output:

	top_scorer	highscore
65	Quinton de Kock	140
33	Jos Buttler	116
71	Rajat Patidar	112
72	Jos Buttler	106
29	Jos Buttler	103
36	KL Rahul	103
25	KL Rahul	103
8	Jos Buttler	100

45	Ruturaj Gaikwad	99
30	Faf du Plessis	96

best_bowling	
Yuzvendra Chahal	5
Rashid Khan	4
T Natarajan	3
Kagiso Rabada	3
Jasprit Bumrah	3
Kuldeep Yadav	3
Avesh Khan	3
Josh Hazlewood	3
Dwayne Bravo	2
Mohsin Khan	2



Interpretation:s

Top_scorer highscore: Jos Buttler is the most consistent batsman. X Frequently contributes highest runs → reliable performer. Teams with consistent top-order batsmen perform better overall. Presence of very high scores (100+) shows matches are often batting dominated. Buttler not just consistent also explosive.

Best_bowling: Chahal and Rashid are key match influencers. They likely take wickets in middle overs, control run flow. “Strong bowling attack is equally important as batting”

5. To analyze Player of the Match trends

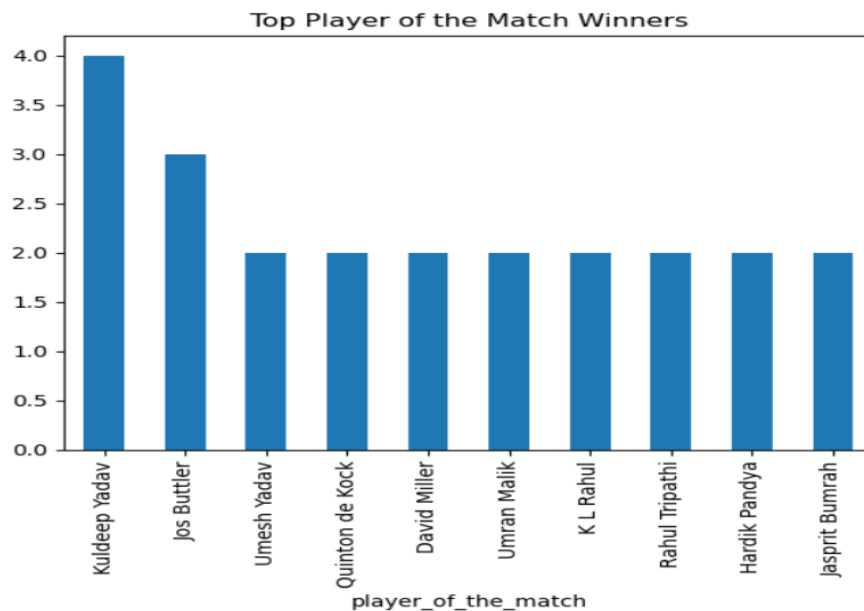
Code:

```
pom = df["player_of_the_match"].value_counts().head(10)

print(pom)
```

Output:

```
player_of_the_match
Kuldeep Yadav      4
Jos Buttler        3
Umesh Yadav        2
Quinton de Kock    2
David Miller       2
Umran Malik        2
K L Rahul          2
Rahul Tripathi     2
Hardik Pandya      2
Jasprit Bumrah     2
```



Interpretation:

Kuldeep Yadav has highest impact performances. Interesting point Top scorer $\neq$ always Player of the Match. “Match winners are not only batsmen — bowlers play crucial roles”.

6. To study venues and match competitiveness

Code:

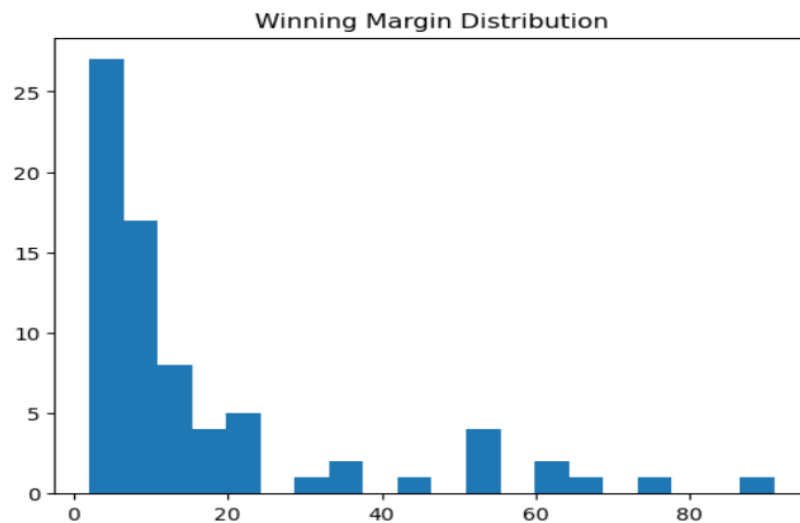
```
venue_wins = df.groupby("venue")["match_winner"].count().sort_values(ascending=False)
print(venue_wins)
```

```
df["margin"] = pd.to_numeric(df["margin"], errors='coerce')
```

```
plt.hist(df["margin"].dropna(), bins=20)
plt.title("Winning Margin Distribution")
plt.show()
```

Output:

venue	
Wankhede Stadium, Mumbai	21
Dr DY Patil Sports Academy, Mumbai	20
Brabourne Stadium, Mumbai	16
Maharashtra Cricket Association Stadium,Pune	13
Eden Gardens, Kolkata	2
Narendra Modi Stadium, Ahmedabad	2



Interpretation:

Most matches concentrated in Mumbai venues. Results may be biased toward pitch conditions. “Venue plays a role in scoring patterns and match outcomes”.

Most matches have low margin (0–20 range). Few matches have very high margins (50+). IPL is mostly competitive, but occasionally dominated by strong teams

❖ Overall Conclusion

The project “IPL Match Data Analysis using EDA” successfully demonstrates how data analysis techniques can be applied to understand real-world sports data from the Indian Premier League. The dataset was cleaned and analyzed to identify key factors influencing match outcomes.

The analysis shows that team performance and player contributions play a major role in determining match results. Gujarat emerged as the most successful team, while players like Jos Buttler and Yuzvendra Chahal showed consistent and impactful performances. The results also indicate that the toss has minimal influence on match outcomes, and both batting first and chasing strategies are equally effective.

Data visualization helped in clearly understanding patterns such as team dominance, player performance, and match competitiveness. Most matches were closely contested, highlighting the competitive nature of the league.

Overall, this project emphasizes the importance of Exploratory Data Analysis in extracting meaningful insights and making data-driven decisions in sports analytics.